

Macro Desktop & the Macro MFE Libraries

One professional, Wells Fargo–branded desktop for the Macro franchise — and a complete toolkit to build for it in days, not months.

● LIVE — UAT & PROD

Macro Desktop is our **OpenFin Workspace** platform for the Macro business — **Rates, Commodities and FX**. It gives traders and sales a single, consistent, branded desktop, and gives engineering teams a paved road to deliver micro frontends (MFEs) onto it quickly and safely.

● IN PRODUCTION

Hyperion apps

Live in PRODUCTION on Macro Desktop, built with the Macro MFE Libraries.

● IN DEVELOPMENT

Hydra apps

In active development on the same platform and shared libraries.

CAPITAL MARKETS UI STRATEGY

A direct on-ramp to the firm-wide Capital Markets UI Strategy

A Capital Markets–wide UI strategy is taking shape along the same lines as our approach, and the Macro UI team is actively participating in it. Because Macro Desktop and the Macro MFE Libraries already follow that model, the MFEs we build today are positioned to integrate into the wider Capital Markets store front with little to no rework.

- ✓ Macro UI team is actively participating in — and will help influence and shape — the strategy.
- ✓ A clear path for the Macro team to build MFEs that drop into the firm-wide store front.
- ✓ Seamless, very low-friction integration into the wider Capital Markets UI.



The Trading Floor workspace. Rates blotter, sales cockpit, order ticket / RFQ, middle office, analytics and admin — independent MFEs arranged in a single saved OpenFin layout.

FOR THE BUSINESS

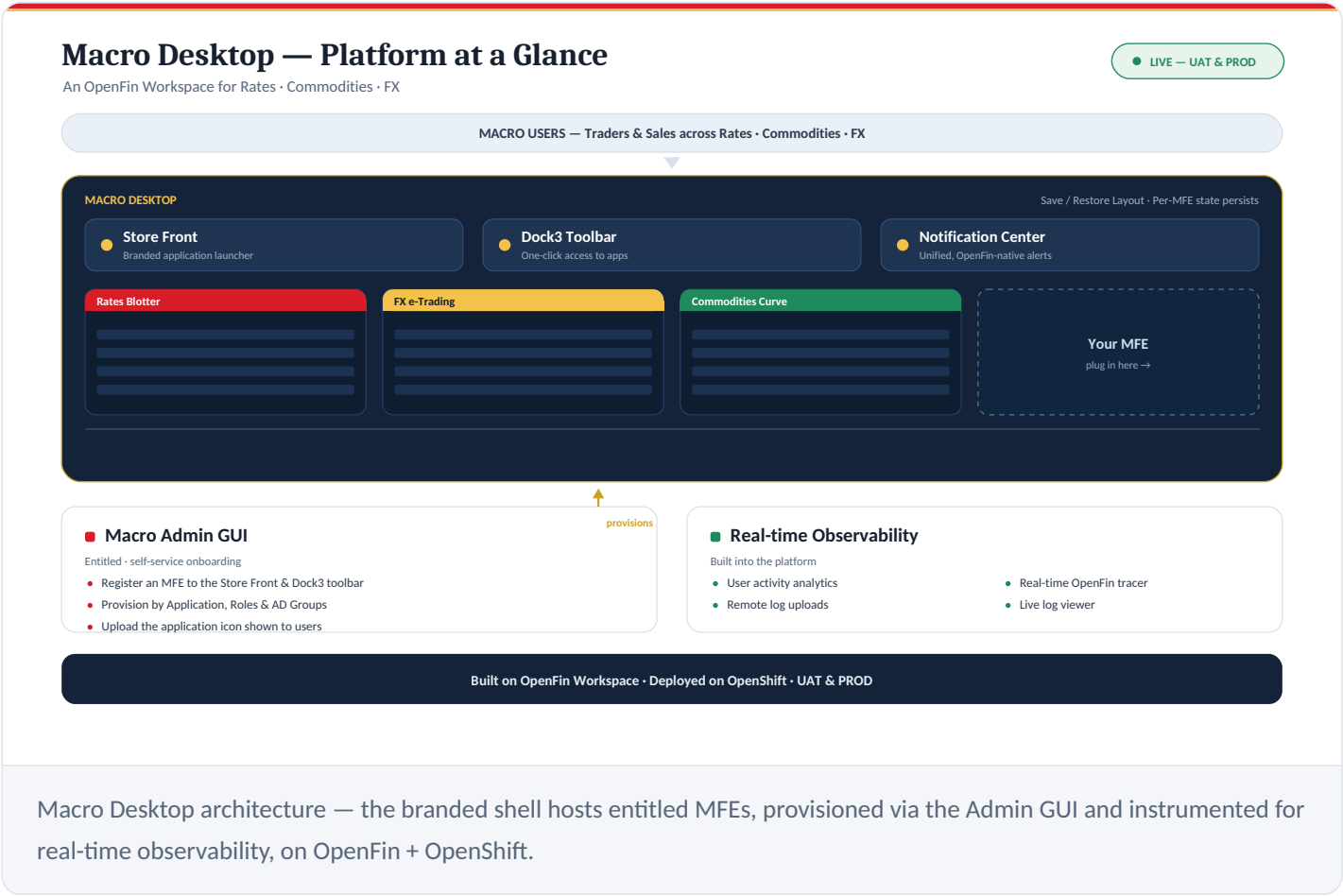
- One branded, professional desktop across Rates, Commodities & FX
- Entitlement-driven — users only see what they're approved for
- Persistent, personalized layouts and preferences
- Real-time analytics and observability across the franchise

FOR ENGINEERING TEAMS

- A scaffolding-first paved road — start compliant, not from a blank repo
- Shared libraries for auth, transport, theming, grids and observability
- Build in **Angular or React** — both first-class
- Self-service onboarding via the Macro Admin GUI

Platform at a glance

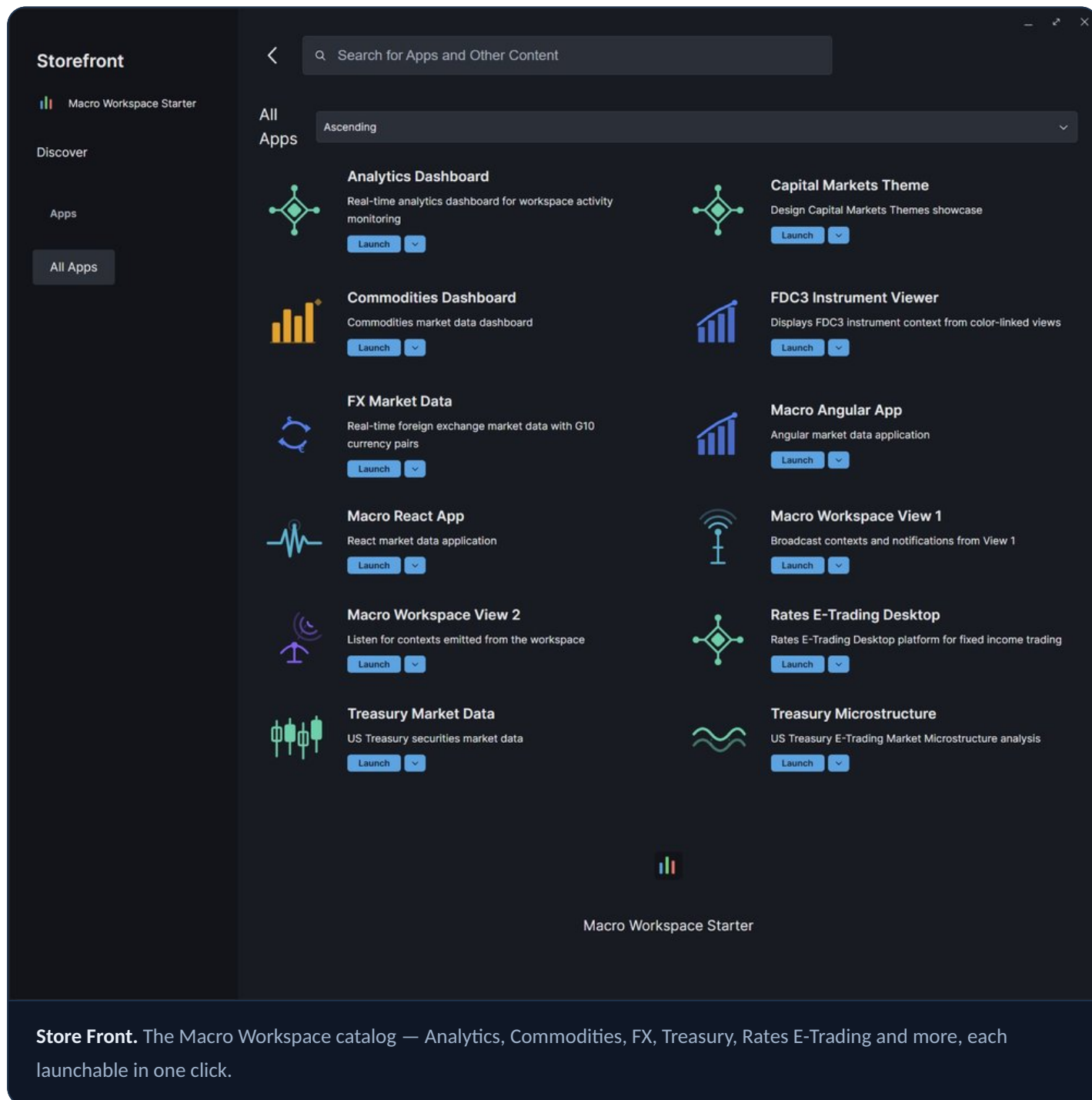
How the desktop, its shell features, onboarding and observability fit together.



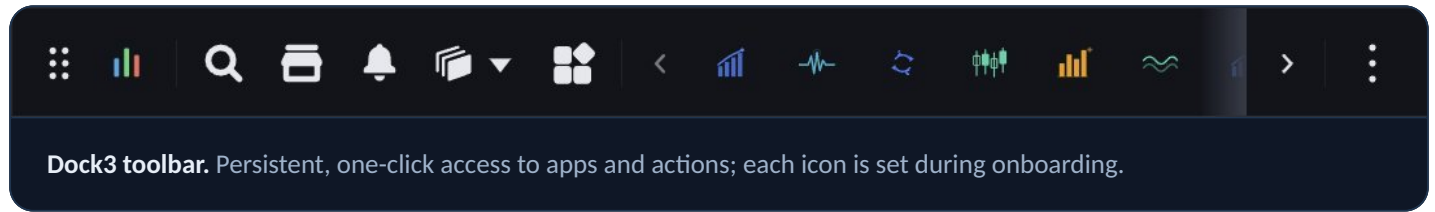
— The branded shell

STORE FRONT

Users discover and launch only the apps they're entitled to — Angular and React MFEs side by side.



DOCK3 TOOLBAR



— Platform capabilities

Branded Store Front

A professional, Wells Fargo-branded launcher where users discover and open the applications they're entitled to.

Dock3 Toolbar

A persistent toolbar giving one-click access to applications and actions.

Notification Center

Centralized, OpenFin-native notifications surfaced consistently across every hosted MFE.

Save / Restore Layout

Users save their full workspace arrangement and restore it across sessions.

Per-MFE State

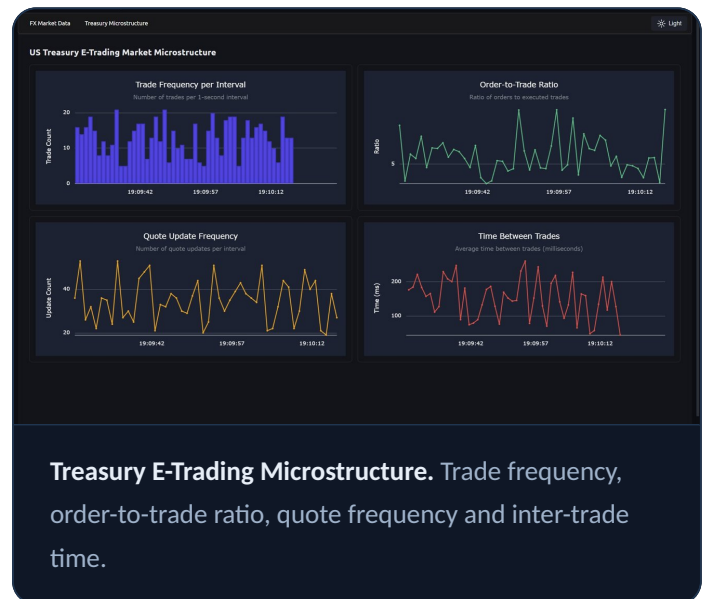
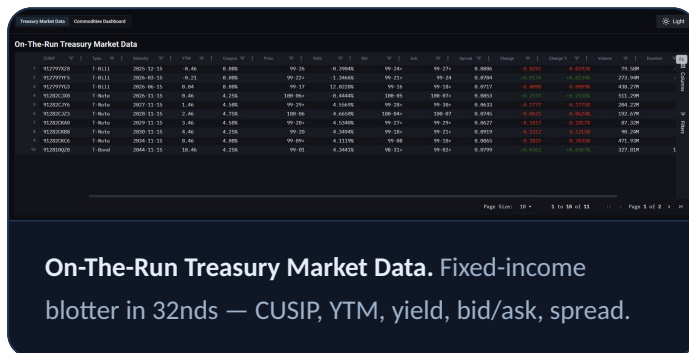
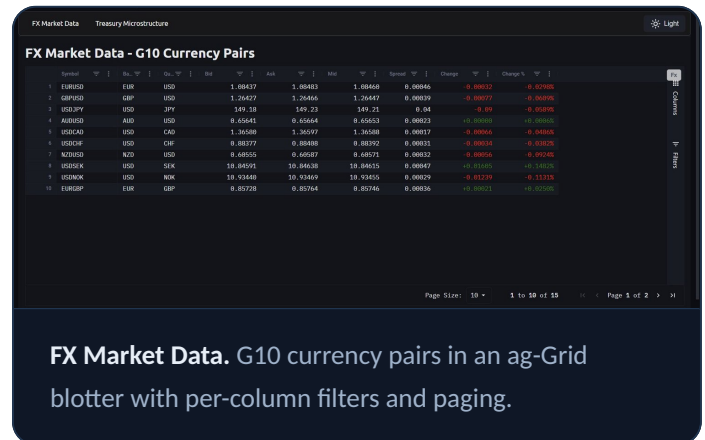
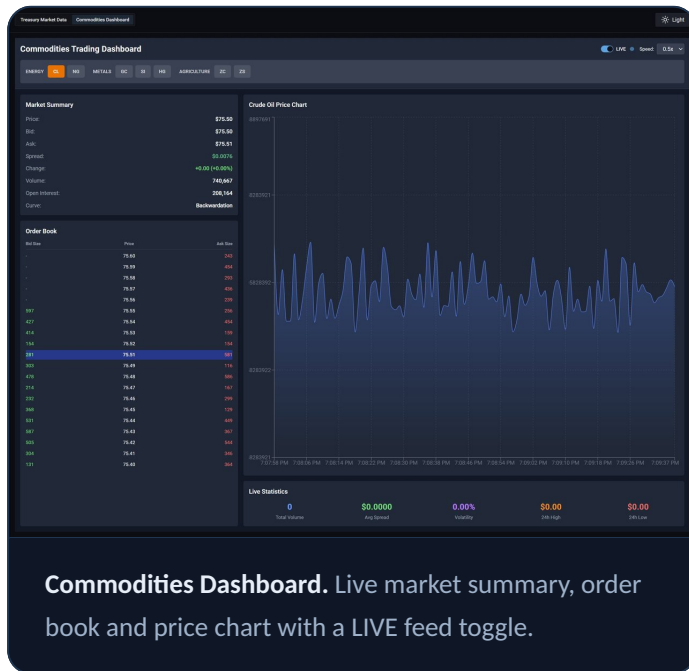
Each MFE persists its own internal state and preferences — every view returns exactly as the user left it.

Real-time Observability

User activity analytics, remote log uploads, a real-time OpenFin tracer and a live log viewer — built in.

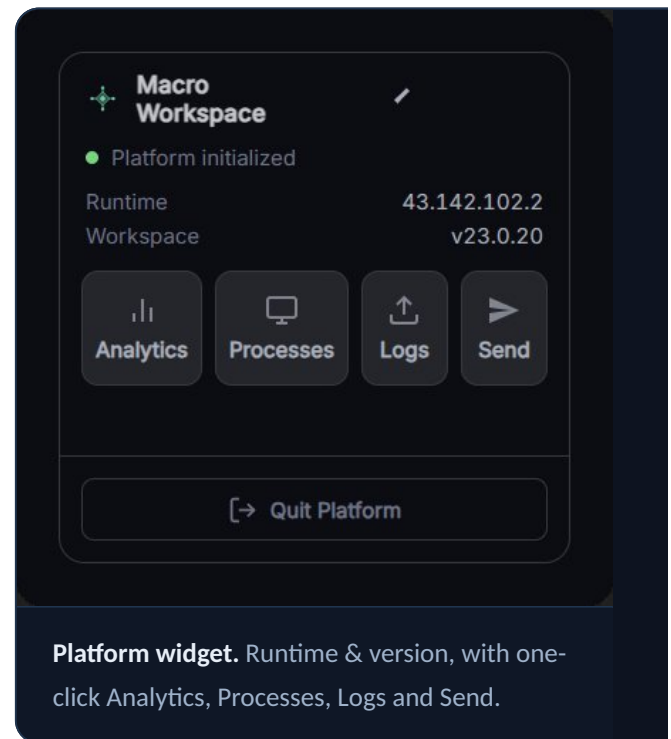
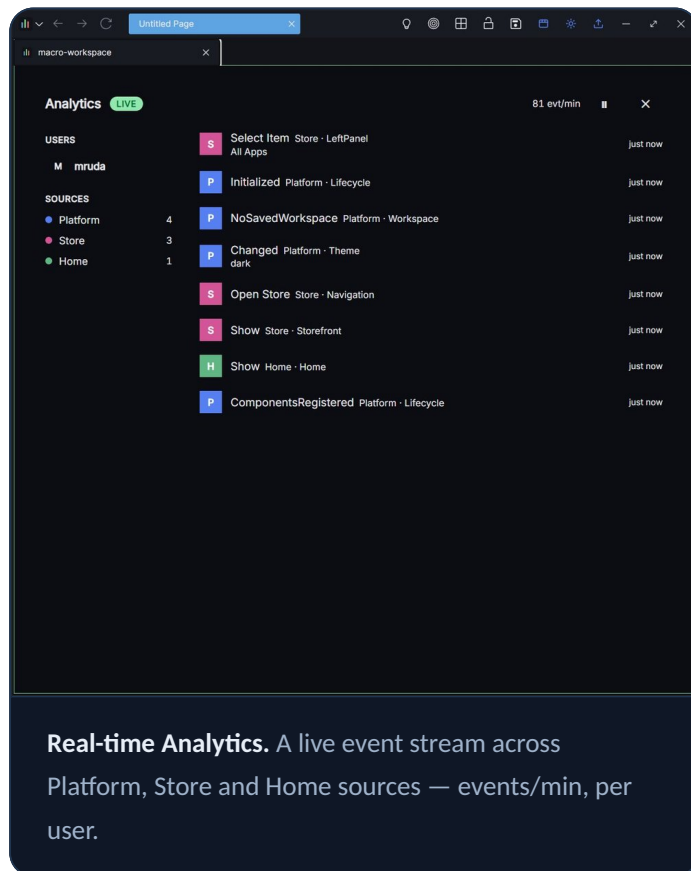
— MFEs in action

A sample of live MFEs already running on the desktop — Angular and React, real-time data, ag-Grid blotters and charts.



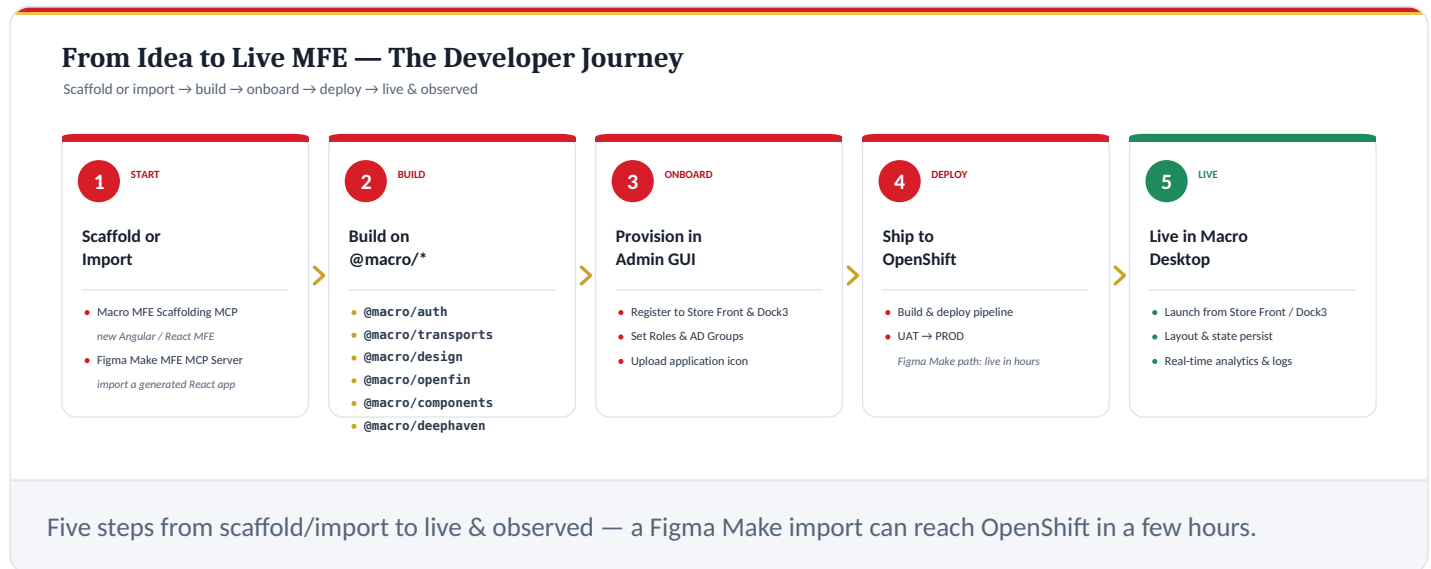
— Built-in observability

Real-time analytics, logs and tracing ship with the platform — no per-app wiring.



— The developer journey

From a design or an idea to a live, branded MFE on the desktop.



01 Scaffold or import

Start a new Angular or React MFE with the **Macro MFE Scaffolding MCP Server**, or bring a **Figma Make-generated React app** into the ecosystem with the **Figma Make MFE MCP Server**.

02 Build on @macro/*

Wire in identity, real-time transport, theming, grids and logging using the shared libraries — focus on application logic, not plumbing.

03 Onboard in the Admin GUI

Register the MFE to the Store Front and Dock3, provision it by Application / Roles / AD Groups, and upload its icon.

04 Ship to OpenShift

Build and deploy through to UAT and PROD. A Figma Make import can be live on OpenShift in a few hours.

05

Live & observed

Users launch from the Store Front and Dock3; layouts and state persist; activity and logs stream into real-time observability.

The Macro MFE Libraries

A complete toolkit for building Angular and React MFEs that are compliant with Macro Desktop from day one — anchored by two MCP servers.

Macro MFE Scaffolding MCP Server

Describe what you're building and get a standards-aligned Angular or React MFE — wired for auth, theming, transport and the platform contract.

Figma Make MFE MCP Server

Import a Figma Make-generated React app directly into the Macro ecosystem and design system, and host it on OpenShift in a few hours.

PACKAGE	WHAT IT PROVIDES	FRAMEWORKS
@macro/auth	Framework-agnostic OAuth 2.0 (PKCE) Auth API with APIGEE / Ping integration; resolves the signed-in user's UID, full name, roles and AD groups.	Angular & React
@macro/design	Design tokens and CSS custom properties; dark / light Macro theme — including ag-Grid theming.	Angular & React
@macro/logger	SLF4J / Log4J-style structured logging.	Angular & React
@macro/openfin	OpenFin utilities: view-state persistence, theming, async notifications and the FDC3 context service.	Angular & React
@macro/transport	Unified messaging transport for AMPS, Solace and WebSocket — RxJS-based, with Angular and React integration.	Angular & React

PACKAGE	WHAT IT PROVIDES	FRAMEWORKS
@macro/utils	Common UI utilities: conflation, price formatters, clipboard and host detection.	Angular & React
@macro/react-components	React component set: ag-Grid wrapper, loading spinner, confirm dialog and more.	React
@macro/components	Angular component set: ag-Grid wrapper, loading spinner, confirm dialog and more.	Angular
@macro/deephaven	Deephaven Core+ JavaScript adapter for ag-Grid, using the ag-Grid Viewport datasource.	Angular & React

— Onboarding & administration

Self-service through the entitled **Macro Admin GUI** — no platform-side code changes.

1. **Register** an MFE to the Store Front and the Dock3 toolbar.
2. **Provision by entitlement** — map the MFE to its Application, required Roles and AD Groups, so it appears only for entitled users.
3. **Brand the listing** — upload a custom application icon shown in both the Store Front and the Dock3 toolbar.

— Getting started

1. **Start** by scaffolding a new MFE, or importing a Figma Make React app, via the MCP servers.
2. **Build** with the `@macro/*` libraries for auth, transport, theming and grids.
3. **Onboard** through the Macro Admin GUI — register, set Roles / AD Groups, upload your icon.
4. **Go live** on OpenShift — UAT, then PROD.

Questions, or ready to onboard an application?

Reach out to **Rubicon** / the **Macro UI team** for more details.